3.1 Measurement Setup

- Connect the output of the pulser to a splitter
- Connect one side of the splitter to the scope
- Connect the other side of the splitter to the input of an amplifier
- PULSER: Set the rise time to be fast (<1 µs)
- PULSER: Set the fall time to be slow (>100 µs)
- Change the amplitude on the pulse generator such that a pulse of ~0.1 V is seen on the scope
- Connect the unipolar output of the amplifier to the other input on the scope
- SCOPE: Configure the scales such that you can easily measure the amplitude of pulses from each input simultaneously
- Set the gain of the amplifier such that the pulses are seen as ~1 V (or as low as possible) on the scope after the amplifier (If you see a bipolar pulse out of the amplifer, please ask GSI for help. It is possible that the pole-zero adjustment is not optimized).

3.2 Measurement Results

Use the scope to measure the pulse amplitude before and after the amplifier. Do this for a range of amplitudes (at least five different amplitudes), by changing the output of the pulser while the amplifier settings remain constant. Record your results in your lab notebook. The gain in the amplifer is the product of coarse gain and fine gain. The fine gain should range from 0.5 to 1.5 (ask GSI if the scale of your amplifer is not correct).

Draw the waveform. For a range of amplitudes set by the pulse generator, increase the amplifier gain until it saturates. What is the saturation voltage of this amplifier?

4 Questions

Include the answers to these questions in your lab notebook.

Section 1.2

Is there a difference between the pulser settings and the values that you measured? What do you think is more accurate: the scope measurements, or the pulser settings? Attempt to explain the discrepancies.

Section 1.3

Will **this** sampling frequency affect your measurement? If the frequency was much lower, how would it affect the measurement of the time (x-axis) and the voltage (y-axis)?

Section 2.2

Explain why you observed each of the amplitudes / waveforms. Describe what is happening at the end of the long cable.

Section 3.1

Plot the output amplitude vs. the input amplitude using a computer and attach the plot in your lab notebook. Do a linear fit to the data. Report the correlation coefficient (the R^2 value in Excel) to evaluate the linearity of this system. Do not include saturation datapoints in this fit.